# 1. Organisation of the Organism

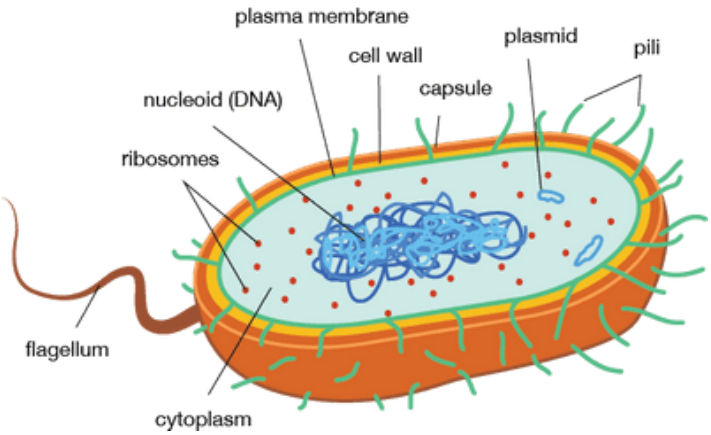
## 1.1. Cell Structure

- All living things are made of cells.
- New cells are produced by the division of existing cells
- All typical eukaryotic cells (multicellular) have:
  - Cell Membrane: controls movement in and out of cells
  - Cytoplasm: where chemical/metabolic reactions take place
  - Nucleus: contains DNA, stores genetic information and controls the activity of the cell
  - Mitochondria: where aerobic respiration happens
  - Ribosome: allows protein synthesis in the cytoplasm
  - Endoplasmic reticulum: Studded with ribosomes, it looks rough under the microscope.
- A typical animal cell (e.g., the liver cell) has all the above.
- Plant cells especially also have:
  - Vacuole: cell sap to keep cell turgid
  - Cell Wall: rigid to hold the shape of the cell, strengthens the cell
  - Chloroplasts: contain chlorophyll, which absorbs light energy for photosynthesis
- A typical plant cell (e.g., the palisade cell) has everything above.

### Prokaryotes

Prokaryotes (Unicellular) DO NOT have mitochondria and a true nucleus.

- One example of a prokaryote is bacteria.
- A bacterial cell only contains a cell wall made of peptidoglycan, cell membrane, cytoplasm, ribosomes, and plasmids.
- It lacks a nucleus and is represented by a circular chromosome of DNA.
- Plasmids are small, circular rings of DNA in the cytoplasm with extra genes outside the chromosomal DNA.




Syllabus 2.1.3: You must be able to identify the cell structures in diagrams and images of plant, animal and bacterial cells

## 1.4. Levels of Organisation

- The division of existing cells produces new cells.

Key Terms

- Cells: Building Blocks of Life
- Tissue: Groups of cells with similar structures working together to perform a shared function
- Organ: Group of tissues working together to perform a specific function
- Organ system: Group of organs with related functions working together to perform body functions.
- Organism: A human

### Specialised Cells

- Specialised Cells have Specific Functions.

| Specialised Cells | Specific Function | Location of Cell |
|---|---|---|
| Ciliated cells | Movement of mucus in the trachea and bronchi | Respiratory Tract, Fallopian Tube, Testes |
| Root Hair cells | Absorption | Roots |
| Palisade Mesophyll cell | Photosynthesis | Leaf Cells |
| Neurones | Conduction of electrical impulses | Everywhere in an organism |
| Red Blood cells | Transport of oxygen | Everywhere in an organism |
| Sperm and Egg cells/ovum (gametes) | For reproduction | In their respective gonads (glands) |

## 1.5. Magnification

The general formula is represented in this ways:

$$Magnification = \frac{\text{size of drawing}}{\text{size of specimen}} = \frac{\text{image}}{\text{actual}} = \frac{I}{A}$$

Other Forms in Magnification Formula

- Actual size = image size ÷ magnification
- Image size = magnification x actual size

Unit Conversions (µm - micrometre)

- 1cm = 10mm
- 1mm = 1000µm
- 1µm = 0.001mm

Magnification does NOT have any units ('x 50' or 'x 5000')

Tip: This comes out frequently in all three papers



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